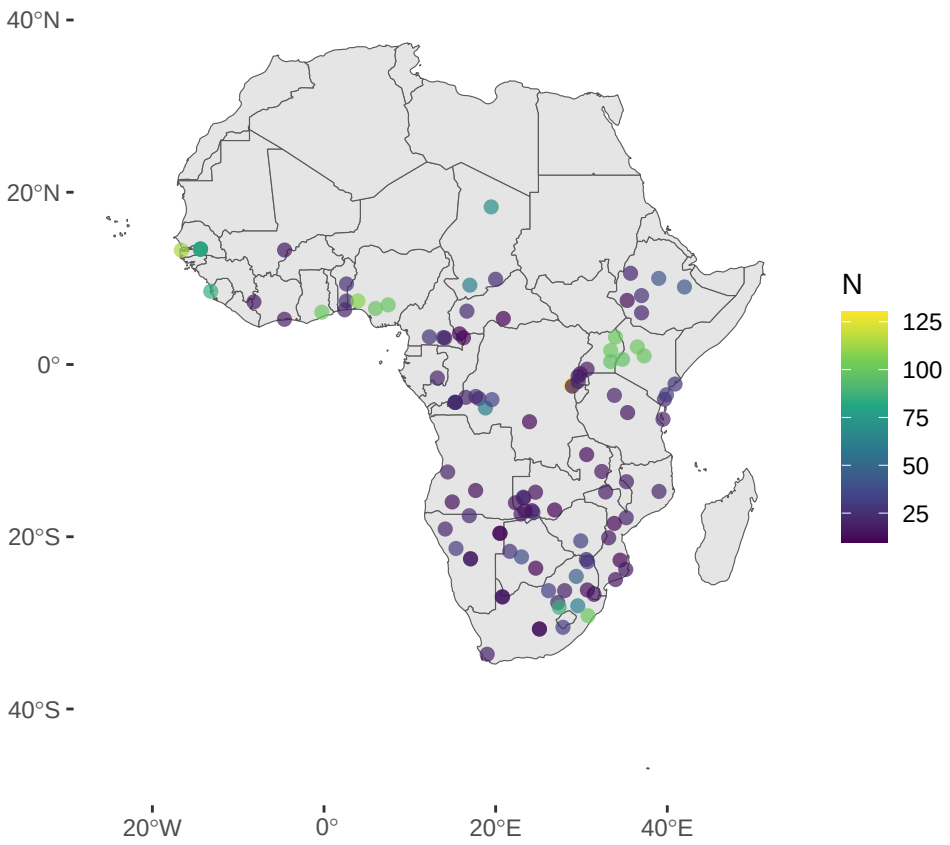
# Bantu African dataset maps

2025-11-04

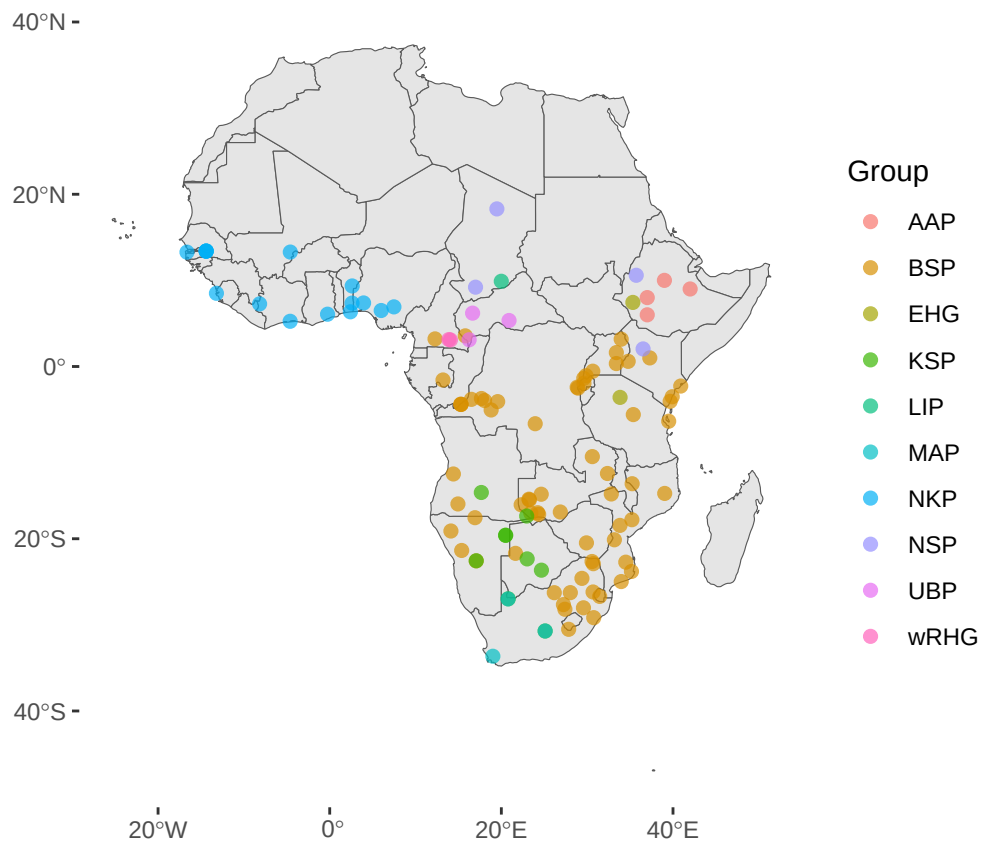
This script combines metadata from the Fortez-Lima et al 2023 paper with our full dataset to create a map showing f3 values for African populations with Mozambique as an African source population and Borneo as an Asian source population.

## Read metadata

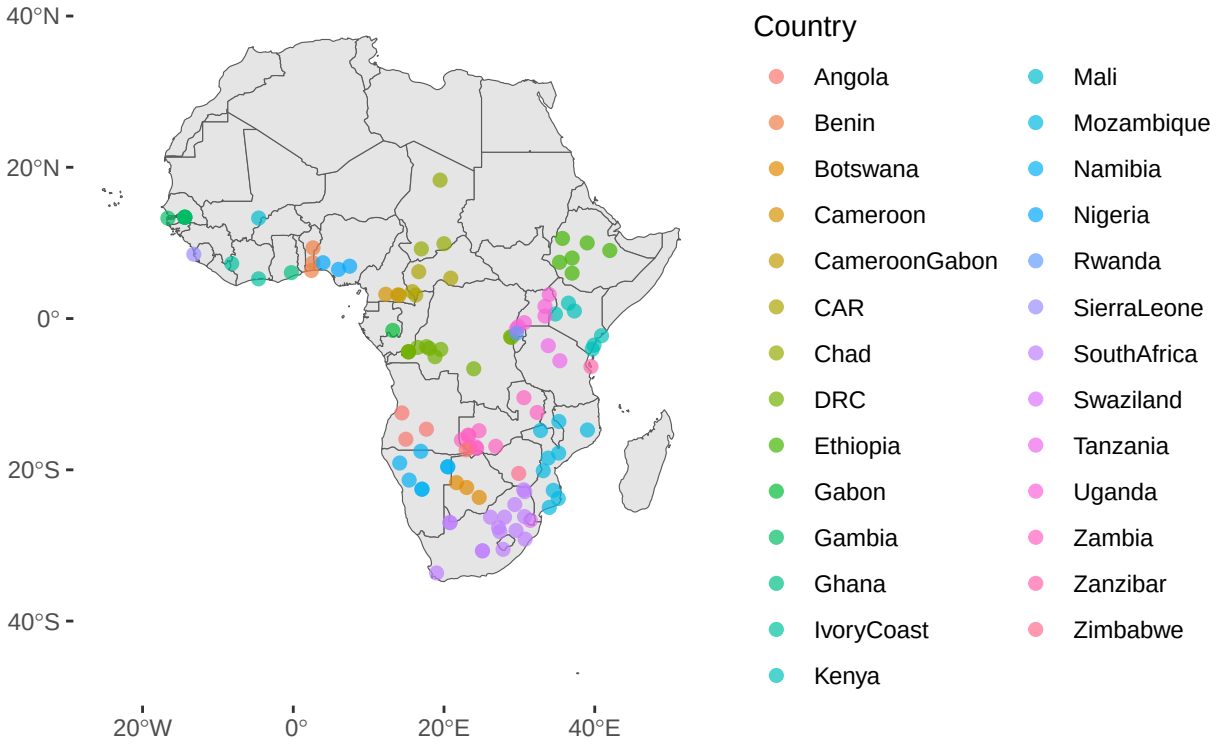
```
# create plot of Africa
africa <- ne_countries(continent = c("Africa"), scale = 50, returnclass = "sf")
# plot by sample size
p2 <- ggplot() +
  geom_sf(data = africa) +
  geom_sf(data = dat2, aes(color = N), size = 2, alpha = 0.7) +
  theme(panel.background = element_rect(fill = "white", color = NA)) +
  scale_color_viridis_c()
p2
```




```
# plot by group
p3 <- ggplot() +
  geom_sf(data = africa) +
  geom_sf(data = dat2, aes(color = Group), size = 2, alpha = 0.7) +
  theme(panel.background = element_rect(fill = "white", color = NA))
p3
```




```
# plot by country
p4 <- ggplot() +
  geom_sf(data = africa) +
  geom_sf(data = dat2, aes(color = Country), size = 2, alpha = 0.7) +
  theme(panel.background = element_rect(fill = "white", color = NA))
p4
```




The metadata plotted above contain 111 populations - the data in the 'Only African' dataset in Fortez-Lima et al 2023.

We have some data data from the 'Full-genotyped' dataset as this contains '81 populations with sample sizes lower than 10 individuals'.

## Read in our full genetic dataset

```
neo_full <- gt_load("../data/intermediate/rbind_data/NEO_full.gt")
length(unique(neo_full$population))
```

```
## [1] 164
```

```
length(unique(dat2$`Population label`))
```

```
## [1] 111
```

Our full dataset contains 164 populations, the main metadata only contain 111.

```
mad_bor_g1000_sum_ld <-
  gt_load("../data/intermediate/rbind_data/mad_bor_g1000_sum_neo_clean_ld.gt")
nrow(mad_bor_g1000_sum_ld)
```

```
## [1] 2744
```

## Get 1000 Genomes populations locations

```
kgpmeta <- kgpmeta
matching_m <- match(mad_bor_g1000_sum_ld$population, kgpmeta$pop)
mad_bor_g1000_sum_ld$Latitude <- ifelse(!is.na(matching_m),
  kgpmeta$lat[matching_m],
  NA
)
mad_bor_g1000_sum_ld$Longitude <- ifelse(!is.na(matching_m),
  kgpmeta$lng[matching_m],
  NA
)

linguistic_groups <- dat2
```

## Additional metadata sheets

Now we read in all sheets in the excel supplementary data to get coordinates for all populations in our full dataset.

Sheet 9 contains data used in the spatially explicit framework, previously published in: https://www.nature.com/articles/s41586-020-2859-7

## Merge metadata

```
# give dat3 and dat4 and dat10 a Country column with value "Unknown"
dat3$Country <- "Unknown"
dat4$Country <- "Unknown"
dat10$Country <- "Unknown"

# remove any duplicates of dat2$population in dat3, dat4, dat9, dat10
dat_2_pops <- dat2$Population
dat3 <- dat3[!dat3$Population %in% dat_2_pops, ]
dat4 <- dat4[!dat4$Population %in% dat_2_pops, ]
dat9 <- dat9[!dat9$Population %in% dat_2_pops, ]
dat10 <- dat10[!dat10$Population %in% dat_2_pops, ]

all_pops <- rbind(dat2, dat3, dat4, dat9, dat10)
dups <- duplicated(all_pops)
# remove duplicates
all_pops <- all_pops[!dups, ]
```

Coordinates are available for 136 populations after merging the metadata.

However, setdiff shows 106 of the populations in neo_full are not in the metadata!

In some cases the mismatch is a naming issue, eg

- Namibia_DamaraKSP in dat$`Population label` is probably “Namibia_Damara” in neo_full)
- SouthAfrica_Coloured in neo_full is SouthAfrica_ColouredAskham in dat$`Population label` - both have 12 individuals

## Fix some naming issues

```
neo_full$population <- ifelse(neo_full$population == "DRC_Bangubangu",
  "DRC_BanguBangu",
  neo_full$population
)
mad_bor_g1000_sum_ld$population <-
  ifelse(mad_bor_g1000_sum_ld$population == "DRC_Bangubangu",
    "DRC_BanguBangu",
    mad_bor_g1000_sum_ld$population
  )

neo_full$population <-
  ifelse(neo_full$population == "SouthAfrica_ColouredWellington",
    "SouthAfrica_Coloured-Wellington",
    neo_full$population
  )
mad_bor_g1000_sum_ld$population <-
  ifelse(mad_bor_g1000_sum_ld$population == "SouthAfrica_ColouredWellington",
    "SouthAfrica_Coloured-Wellington",
    mad_bor_g1000_sum_ld$population
  )

neo_full$population <-
  ifelse(neo_full$population == "SouthAfrica_ColouredColesberg",
    "SouthAfrica_Coloured-Colesberg",
    neo_full$population
  )
mad_bor_g1000_sum_ld$population <-
  ifelse(mad_bor_g1000_sum_ld$population == "SouthAfrica_ColouredColesberg",
    "SouthAfrica_Coloured-Colesberg",
    mad_bor_g1000_sum_ld$population
  )

neo_full$population <- ifelse(neo_full$population == "Namibia_Damara",
  "Namibia_Damara-KSP",
  neo_full$population
)
mad_bor_g1000_sum_ld$population <-
  ifelse(mad_bor_g1000_sum_ld$population == "Namibia_Damara",
    "Namibia_Damara-KSP",
    mad_bor_g1000_sum_ld$population
  )

neo_full$population <- ifelse(neo_full$population == "SouthAfrica_Coloured",
  "SouthAfrica_Coloured-Askham",
  neo_full$population
)
mad_bor_g1000_sum_ld$population <-
  ifelse(mad_bor_g1000_sum_ld$population == "SouthAfrica_Coloured",
    "SouthAfrica_Coloured-Askham",
    mad_bor_g1000_sum_ld$population
  )
```

## Add country, latitude, and longitude

```
matching <- match(mad_bor_g1000_sum_ld$population, all_pops$Population)
mad_bor_g1000_sum_ld$Country <-
  all_pops$Country[matching]
# add lat and long to mad_bor_g1000_sum_ld
mad_bor_g1000_sum_ld$Latitude <- ifelse(is.na(mad_bor_g1000_sum_ld$Latitude),
  all_pops$Latitude[matching],
  mad_bor_g1000_sum_ld$Latitude
)
mad_bor_g1000_sum_ld$Longitude <- ifelse(is.na(mad_bor_g1000_sum_ld$Longitude),
  all_pops$Longitude[matching],
  mad_bor_g1000_sum_ld$Longitude
)
```

```
gt_save(
  mad_bor_g1000_sum_ld,
  "../data/intermediate/rbind_data/mad_bor_g1000_sum_neo_clean_coords.gt"
)
```

```
##
## gen_tibble saved to ../data/intermediate/rbind_data/mad_bor_g1000_sum_neo_clean_coords.gt

## using FBM RDS: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/mad_bor_g

## with FBM backing file: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/m

## make sure that you do NOT delete those files!

## to reload the gen_tibble in another session, use:

## gt_load('../data/intermediate/rbind_data/mad_bor_g1000_sum_neo_clean_coords.gt')

## [1] "../data/intermediate/rbind_data/mad_bor_g1000_sum_neo_clean_coords.gt"
## [2] "/home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/mad_bor_g1000_sum_n
## [3] "/home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/mad_bor_g1000_sum_n
```

## Filter to populations in neo_full

```
pop <- c(
  "KHV", "BEB", "PJL", "CHS",
  "Borneo",
  "Sumba", "Timor",
  "Mikea", "Vezo",
  "Temoro", "Diego",
  "Merina",
  "Betsimisaraka",
```

```
  "Sakalava_Tsimihety",
  "Betsileo"
)
mad_bor_g1000_sum_ld <- mad_bor_g1000_sum_ld %>% filter(!population %in% pop)
nrow(mad_bor_g1000_sum_ld)
```

```
## [1] 2060
```

```
length(unique(mad_bor_g1000_sum_ld$population))
```

```
## [1] 164
```

```
length(
  mad_bor_g1000_sum_ld %>%
    filter(is.na(Latitude)) %>%
    select(population) %>%
    unique() %>%
    nrow()
)
```

```
## [1] 1
```

Still 98 populations do not contain location metadata.

```
mad_bor_g1000_sum_ld %>%
  group_by(population) %>%
  filter(is.na(Latitude)) %>%
  summarize(count = n()) %>%
  filter(count > 1)
```

```
## # A tibble: 57 x 2
##    population         count
##    <chr>              <int>
##  1 Angola_Sekele          7
##  2 Angola_Vasekele        4
##  3 Botswana_Kalanga       3
##  4 Botswana_Kgalagadi     6
##  5 Botswana_TswanaBot     6
##  6 CAR_Nzakara            8
##  7 CAR_Zande              6
##  8 DRC_BanguBangu         3
##  9 DRC_Hungan             3
## 10 DRC_Kuba               3
## # i 47 more rows
```

We do not calculate f3's for populations with only one individual, so we have 57 populations for which we have calculated f3's that do not have location metadata.

```
mad_bor_g1000_sum_ld %>%
  group_by(population) %>%
  filter(is.na(Latitude)) %>%
  summarize(count = n()) %>%
  filter(count > 1)
```

```
## # A tibble: 56 x 2
##    population         count
##    <chr>              <int>
##  1 Angola_Sekele          7
##  2 Angola_Vasekele        4
##  3 Botswana_Kalanga       3
##  4 Botswana_Kgalagadi     6
##  5 Botswana_TswanaBot     6
##  6 CAR_Nzakara            8
##  7 CAR_Zande              6
##  8 DRC_BanguBangu         3
##  9 DRC_Hungan             3
## 10 DRC_Kuba               3
## # i 46 more rows
```

## Plot populations with genetic data and coordinates

```
mad_is_na <- mad_bor_g1000_sum_ld %>%
  filter(is.na(Latitude))

# sort by alphabetical order of population
mad_is_na <- mad_is_na %>%
  arrange(population)

length(unique(mad_is_na$population)) # 99 populations without latitude
```

```
## [1] 97
```

```
mad_bor_g1000_sum_ld_no_na <- mad_bor_g1000_sum_ld %>%
  filter(!is.na(Latitude))

mad_bor_g1000_sum_ld_no_na <- gt_add_sf(mad_bor_g1000_sum_ld_no_na,
  coords = c("Longitude", "Latitude"), crs = st_crs(4326)
)

africa <- ne_countries(continent = c("Africa"), scale = 50, returnclass = "sf")
```

## Read in f3's

```
adm_f3 <- read.table("../results/s06_b_f_stats/f3_results_all_combinations",
  sep = "\t", header = TRUE,
  stringsAsFactors = FALSE
)

adm_f3_african <- adm_f3 %>%
  filter(Asian == "Borneo")
```

## Rename f3 populations as above

```
adm_f3_african$African <- ifelse(adm_f3_african$African == "DRC_Bangubangu",
  "DRC_BanguBangu",
  adm_f3_african$African
)

adm_f3_african$African <-
  ifelse(adm_f3_african$African == "SouthAfrica_ColouredWellington",
    "SouthAfrica_Coloured-Wellington",
    adm_f3_african$African
  )

adm_f3_african$African <-
  ifelse(adm_f3_african$African == "SouthAfrica_ColouredColesberg",
    "SouthAfrica_Coloured-Colesberg",
    adm_f3_african$African
  )

adm_f3_african$African <- ifelse(adm_f3_african$African == "Namibia_Damara",
  "Namibia_Damara-KSP",
  adm_f3_african$African
)

adm_f3_african$African <-
  ifelse(adm_f3_african$African == "SouthAfrica_Coloured",
    "SouthAfrica_Coloured-Askham",
    adm_f3_african$African
  )

adm_f3_african$African <-
  ifelse(adm_f3_african$African == "Mandinka, The Gambia",
    "GWD",
    adm_f3_african$African
  )

adm_f3_african$African <-
  ifelse(adm_f3_african$African == "Yoruba, Nigeria",
    "YRI",
    adm_f3_african$African
  )

adm_f3_african$African <-
  ifelse(adm_f3_african$African == "Mende, Sierra Leone",
    "MSL",
    adm_f3_african$African
  )

adm_f3_african$African <-
  ifelse(adm_f3_african$African == "Esan, Nigeria",
    "ESN",
    adm_f3_african$African
  )
```

```
adm_f3_african$African <-
  ifelse(adm_f3_african$African == "Luhya, Kenya",
    "LWK",
    adm_f3_african$African
  )
```

## Add f3's to mad_bor_g1000_sum_ld_no_na

```
# create a subset for each Malagasy population
for (i in unique(adm_f3_african$Malagasy)) {
  malagasy_pop <- i
  subset <- paste0("adm_f3_african_", malagasy_pop)
  assign(subset, adm_f3_african %>% filter(Malagasy == malagasy_pop))
}

# list all the subsets
subset_list <- ls(pattern = "adm_f3_african_")

for (i in subset_list) {
  malagasy_pop <- str_replace(i, "adm_f3_african_", "")
  f3 <- paste0("f3_", malagasy_pop)
  se <- paste0("se_", malagasy_pop)
  dataset <- get(i)
  mad_bor_g1000_sum_ld_no_na <- mad_bor_g1000_sum_ld_no_na %>%
    mutate(!!f3 := dataset$est[match(population, dataset$African)]) %>%
    mutate(!!se := dataset$se[match(population, dataset$African)])
}

mad_bor_g1000_sum_ld_no_na <- mad_bor_g1000_sum_ld_no_na %>%
  filter(!is.na(f3_Betsileo))
```

## Add linguistic group

```
mad_bor_g1000_sum_ld_no_na <- mad_bor_g1000_sum_ld_no_na %>%
  mutate(linguistic_group = linguistic_groups$Group[match(
    mad_bor_g1000_sum_ld_no_na$population,
    linguistic_groups$`Population label`
  )])

# add Mozambique_Ronga linguistic_group as BSP
mad_bor_g1000_sum_ld_no_na$linguistic_group <-
  ifelse(mad_bor_g1000_sum_ld_no_na$population == "Mozambique_Ronga",
    "BSP",
    mad_bor_g1000_sum_ld_no_na$linguistic_group
  )

# add Zambia_Tumbuka linguistic_group as BSP
```

```
mad_bor_g1000_sum_ld_no_na$linguistic_group <-
  ifelse(mad_bor_g1000_sum_ld_no_na$population == "Zambia_Tumbuka",
    "BSP",
    mad_bor_g1000_sum_ld_no_na$linguistic_group
  )

# add GWD, MSL, ESN, and YRI as NKP
mad_bor_g1000_sum_ld_no_na$linguistic_group <-
  ifelse(mad_bor_g1000_sum_ld_no_na$population %in% c(
    "GWD",
    "MSL",
    "ESN",
    "YRI"
  ),
  "NKP",
  mad_bor_g1000_sum_ld_no_na$linguistic_group
  )
# add LWK as BSP
mad_bor_g1000_sum_ld_no_na$linguistic_group <-
  ifelse(mad_bor_g1000_sum_ld_no_na$population == "LWK",
    "BSP",
    mad_bor_g1000_sum_ld_no_na$linguistic_group
  )
```

```
# African linguistic groups:

# Bantu-speaking_populations (BSP)
# Niger-Kongo-speaking_populations (NKP)
# Southern_African_Khoe-San_populations (KSP)
# Mixed_ancestry_populations (MAP)
# Ubangi-speaking_populations (UBP)

# replace linguistic_group acronym with full name
mad_bor_g1000_sum_ld_no_na$linguistic_group <-
  ifelse(mad_bor_g1000_sum_ld_no_na$linguistic_group == "BSP",
    "Bantu",
    mad_bor_g1000_sum_ld_no_na$linguistic_group
  )

mad_bor_g1000_sum_ld_no_na$linguistic_group <-
  ifelse(mad_bor_g1000_sum_ld_no_na$linguistic_group == "NKP",
    "Niger-Kongo",
    mad_bor_g1000_sum_ld_no_na$linguistic_group
  )

mad_bor_g1000_sum_ld_no_na$linguistic_group <-
  ifelse(mad_bor_g1000_sum_ld_no_na$linguistic_group == "KSP",
    "Khoe-San",
    mad_bor_g1000_sum_ld_no_na$linguistic_group
  )
mad_bor_g1000_sum_ld_no_na$linguistic_group <-
  ifelse(mad_bor_g1000_sum_ld_no_na$linguistic_group == "MAP",
    "Mixed ancestry",
```

```
    mad_bor_g1000_sum_ld_no_na$linguistic_group
  )

mad_bor_g1000_sum_ld_no_na$linguistic_group <-
  ifelse(mad_bor_g1000_sum_ld_no_na$linguistic_group == "UBP",
    "Ubangi",
    mad_bor_g1000_sum_ld_no_na$linguistic_group
  )
```

```
mad_bor_g1000_sum_ld_no_na$bantu_non_bantu <-
  ifelse(mad_bor_g1000_sum_ld_no_na$linguistic_group == "Bantu",
    "Bantu",
    "Non-Bantu"
  )

mad_bor_g1000_sum_ld_no_na %>%
  select(population, bantu_non_bantu) %>%
  distinct() %>%
  group_by(bantu_non_bantu) %>%
  summarize(count = n())
```

```
## Simple feature collection with 2 features and 2 fields
## Geometry type: MULTIPOINT
## Dimension:     XY
## Bounding box:  xmin: -16.57903 ymin: -33.64328 xmax: 39.49952 ymax: 13.45488
## Geodetic CRS:  WGS 84
## # A tibble: 2 x 3
##   bantu_non_bantu count                                              geometry
##   <chr>           <int>                                    <MULTIPOINT [°]>
## 1 Bantu              47 ((15.79443 3.559467), (35.37383 -5.582431), (39.49952 -~
## 2 Non-Bantu          17 ((-16.57903 13.45488), (-13.23 8.48), (3.92 7.4), (7.48~
```

## Plot f3's on the map: All groups

```
# list all the subsets
subset_list <- ls(pattern = "adm_f3_african_")

plot_maps <- function(subset_list, mad_bor_g1000_sum_ld_no_na, africa) {
  counter <- 1
  plot_list <- list()

  for (i in subset_list) {
    malagasy_pop <- str_replace(i, "adm_f3_african_", "")
    f3 <- paste0("f3_", malagasy_pop)
    se <- paste0("se_", malagasy_pop)

    minimum_f3 <- mad_bor_g1000_sum_ld_no_na %>%
      slice_min(.data[[f3]])
    minval <- unique(as.numeric(minimum_f3[[f3]]))
```

```r
upper <- minval + (10 * unique(as.numeric(minimum_f3[[se]])))

maximum_f3 <- mad_bor_g1000_sum_ld_no_na %>%
  slice_max(.data[[f3]])
maxval <- unique(as.numeric(maximum_f3[[f3]]))

this_plot <- ggplot() +
  geom_sf(data = africa) +
  geom_sf(
    data = mad_bor_g1000_sum_ld_no_na,
    aes(
      colour = .data[[f3]],
      shape  = bantu_non_bantu
    ),
    alpha = 0.3, size = 2
  ) +
  scale_color_gradientn(
    colors = rev(viridis::viridis(10)),
    na.value = "grey80",
    limits = c(minval, maxval), # Using the calculated minval
    # Mapping: Purple at min, Yellow at -0.009, End of scale at 0
    values = scales::rescale(c(seq(minval, upper, length.out = 9), maxval)),
    guide = guide_colorbar(barheight = 6),
    oob = scales::squish
  ) +
  coord_sf(crs = st_crs(4326)) +
  theme_minimal() +
  labs(colour = "f3", shape = "Linguistic group") +
  guides(alpha = "none", size = "none") +
  ggtitle(malagasy_pop) +
  theme(
    legend.position.inside = c(0.15, 0.31),
    legend.title = element_text(size = 10),
    legend.text = element_text(size = 8),
    legend.background = element_rect(fill = "white", color = "grey80"),
    legend.spacing = unit(0.1, "cm")
  )

mins <- mad_bor_g1000_sum_ld_no_na %>%
  select(population, .data[[f3]], .data[[se]]) %>%
  distinct() %>%
  slice_min(.data[[f3]], n = 15)

mad_bor_g1000_sum_ld_no_na$population <-
  gsub("_", " ", mad_bor_g1000_sum_ld_no_na$population)
mins$population <- gsub("_", " ", mins$population)

se_plot <- mad_bor_g1000_sum_ld_no_na %>%
  filter(population %in% mins$population) %>%
  ggplot(aes(
    x = .data[[f3]],
    y = reorder(population, .data[[f3]], decreasing = TRUE)
  )) +
```

```
      geom_point() +
      ggtitle(malagasy_pop) +
      geom_errorbar(
        aes(
          xmin = .data[[f3]] + (.data[[se]]),
          xmax = .data[[f3]] - (.data[[se]])
        ),
        width = 0.2
      ) +
      labs(x = "f3", y = NULL) +
      theme_minimal() +
      theme(
        legend.position = "none",
        axis.line = element_blank(),
        panel.grid.major = element_blank(),
        panel.grid.minor = element_blank(),
        panel.background = element_rect(fill = "white")
      )

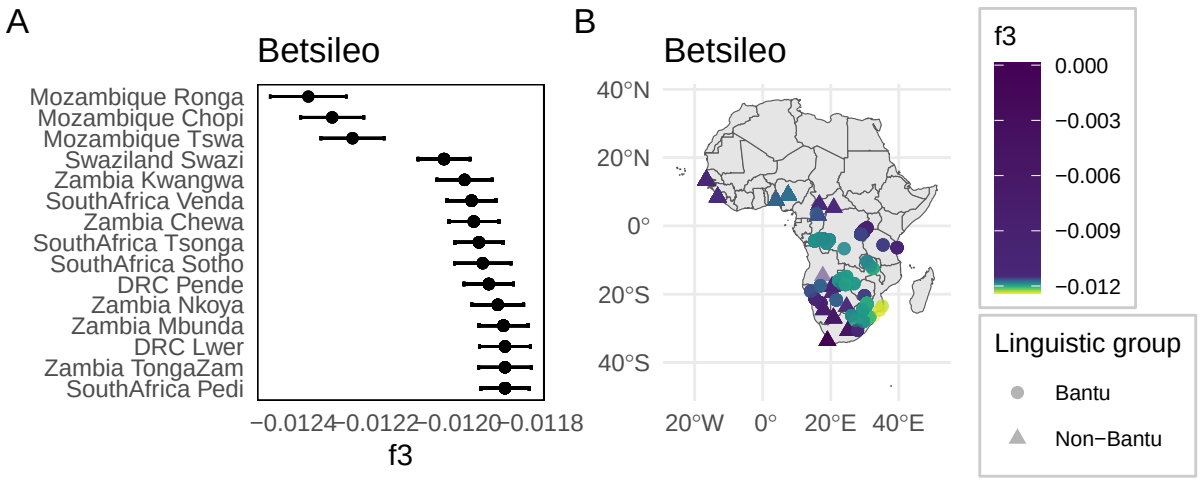
    combined_plot <- wrap_plots(list(se_plot, this_plot), ncol = 2) +
      plot_annotation(
        tag_levels = "A"
      )
    plot_list[[counter]] <- combined_plot
    counter <- counter + 1
  }

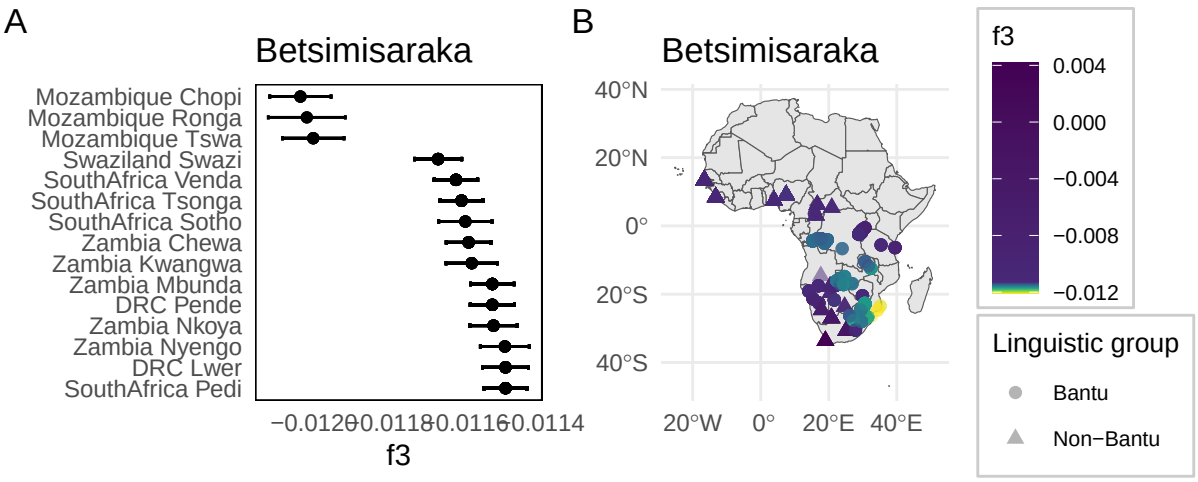
  return(plot_list)
}

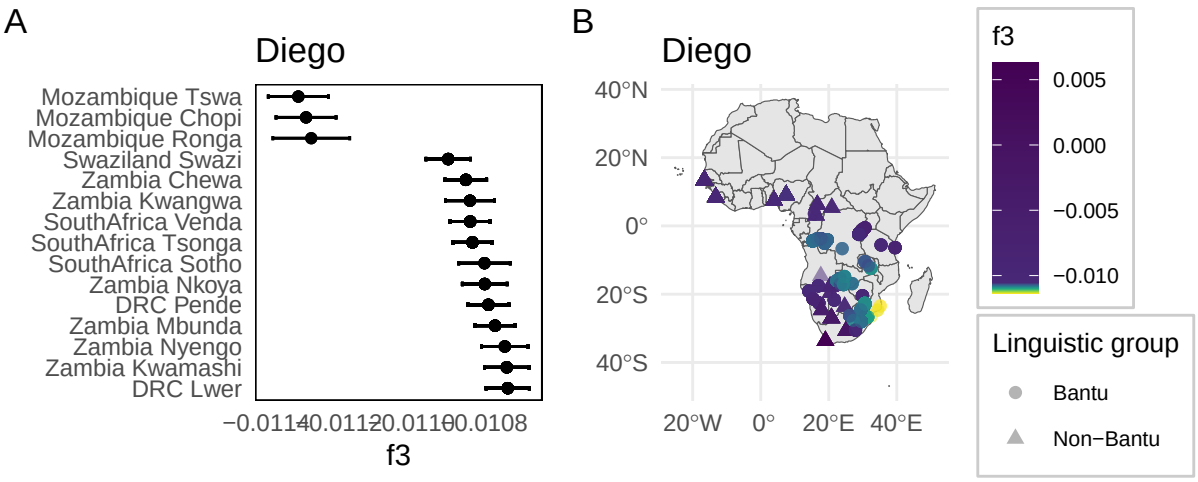
res_afr <- plot_maps(subset_list, mad_bor_g1000_sum_ld_no_na, africa)
```

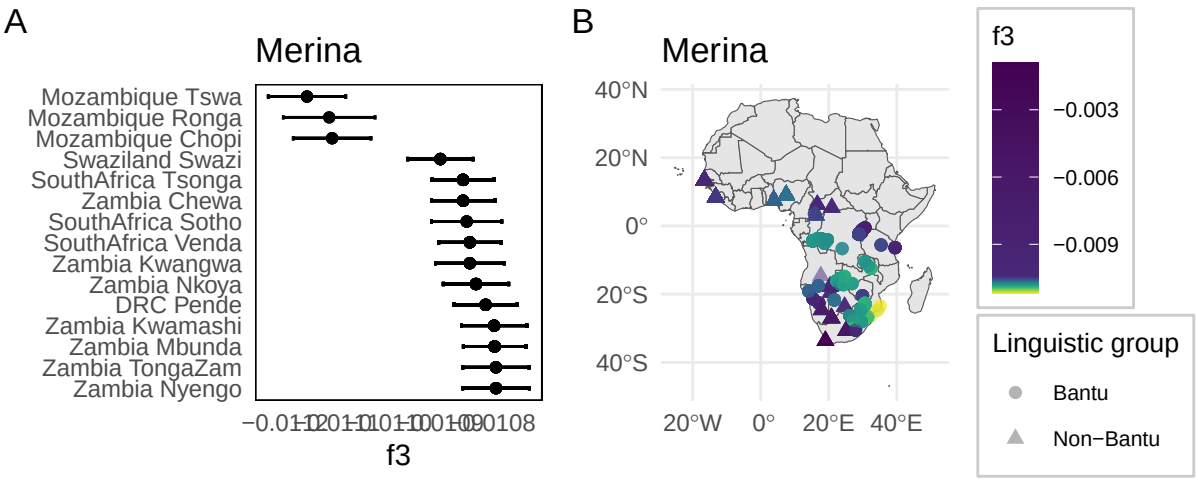
```
## Warning: Use of .data in tidyselect expressions was deprecated in tidyselect 1.2.0.
## i Please use `all_of(var)` (or `any_of(var)`) instead of `.data[[var]]`
## This warning is displayed once per session.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```

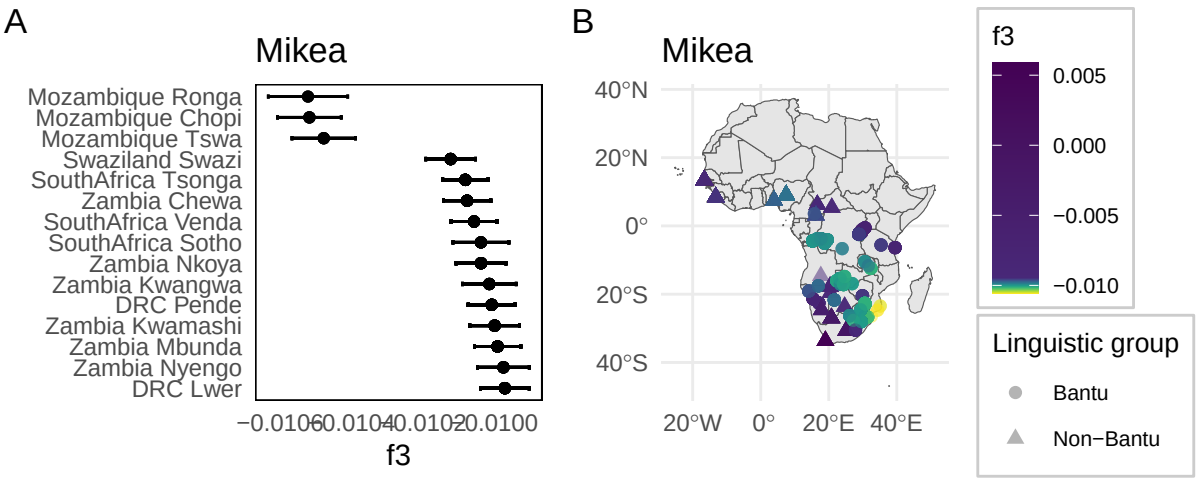
```
for (p in res_afr) {
  print(p)
  ggsave(paste0(res_dir, "f3_all_african_pops_", p$labels$title, ".jpg"),
    width = 10, height = 5, dpi = 300
  )
}
```

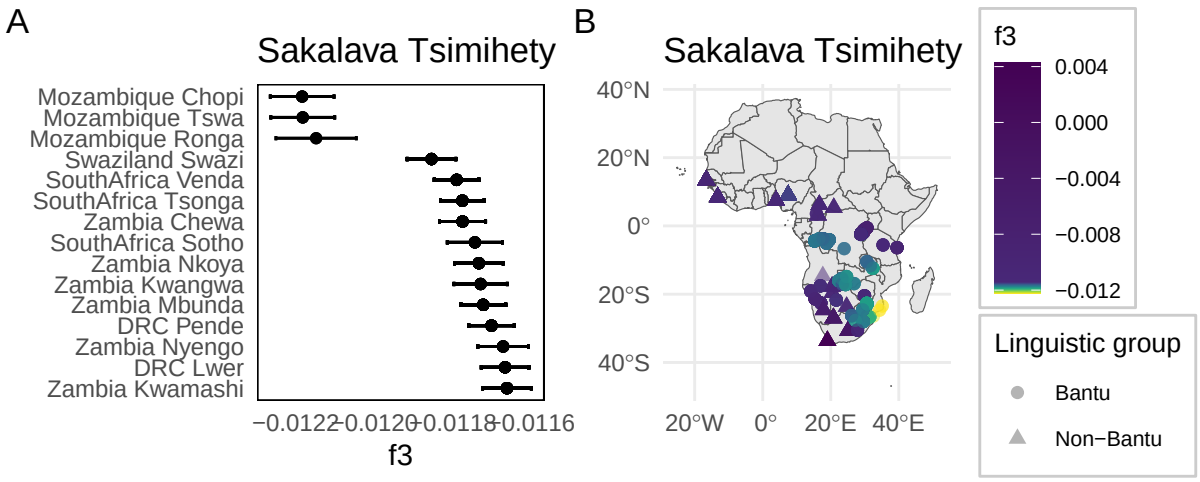

A
Betsileo
Mozambique Ronga
Mozambique Chopi
Mozambique Tswa
Swaziland Swazi
Zambia Kwangwa
SouthAfrica Venda
Zambia Chewa
SouthAfrica Tsonga
SouthAfrica Sotho
DRC Pende
Zambia Nkoya
Zambia Mbunda
DRC Lwer
Zambia TongaZam
SouthAfrica Pedi
−0.0124
−0.0122
−0.0120
−0.0118
f3
B
Betsileo
40°N
20°N
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20°S
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20°W
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20°E
40°E
f3
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−0.003
−0.006
−0.009
−0.012
Linguistic group
Bantu
Non−Bantu

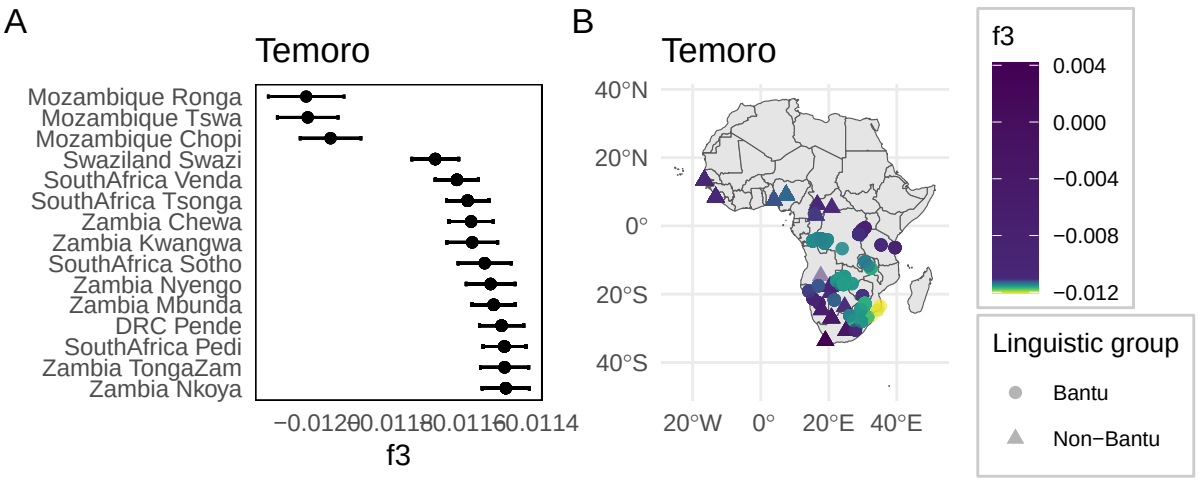

A
Betsimisaraka
Mozambique Chopi
Mozambique Ronga
Mozambique Tswa
Swaziland Swazi
SouthAfrica Venda
SouthAfrica Tsonga
SouthAfrica Sotho
Zambia Chewa
Zambia Kwangwa
Zambia Mbunda
DRC Pende
Zambia Nkoya
Zambia Nyengo
DRC Lwer
SouthAfrica Pedi
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f3
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Betsimisaraka
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f3
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−0.008
−0.012
Linguistic group
Bantu
Non−Bantu

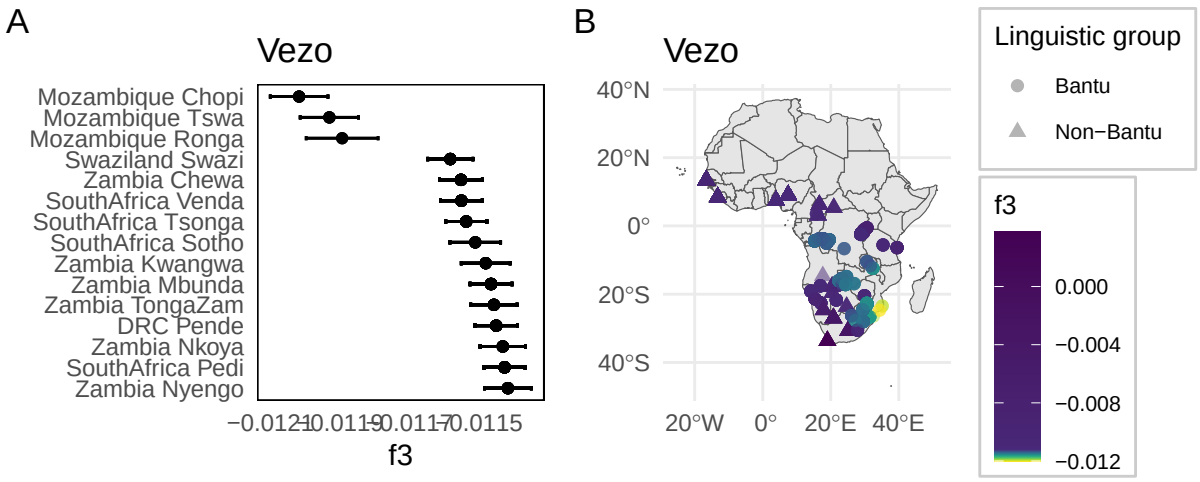

A
Diego
Mozambique Tswa
Mozambique Chopi
Mozambique Ronga
Swaziland Swazi
Zambia Chewa
Zambia Kwangwa
SouthAfrica Venda
SouthAfrica Tsonga
SouthAfrica Sotho
Zambia Nkoya
DRC Pende
Zambia Mbunda
Zambia Nyengo
Zambia Kwamashi
DRC Lwer
−0.0114
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f3
B
Diego
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20°W
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20°E
40°E
f3
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0.000
−0.005
−0.010
Linguistic group
Bantu
Non−Bantu


A
Merina
Mozambique Tswa
Mozambique Ronga
Mozambique Chopi
Swaziland Swazi
SouthAfrica Tsonga
Zambia Chewa
SouthAfrica Sotho
SouthAfrica Venda
Zambia Kwangwa
Zambia Nkoya
DRC Pende
Zambia Kwamashi
Zambia Mbunda
Zambia TongaZam
Zambia Nyengo
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f3
B
Merina
40°N
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f3
−0.003
−0.006
−0.009
Linguistic group
Bantu
Non−Bantu


A
Mikea
Mozambique Ronga
Mozambique Chopi
Mozambique Tswa
Swaziland Swazi
SouthAfrica Tsonga
Zambia Chewa
SouthAfrica Venda
SouthAfrica Sotho
Zambia Nkoya
Zambia Kwangwa
DRC Pende
Zambia Kwamashi
Zambia Mbunda
Zambia Nyengo
DRC Lwer
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−0.010
Linguistic group
Bantu
Non−Bantu


A
Sakalava Tsimihety
Mozambique Chopi
Mozambique Tswa
Mozambique Ronga
Swaziland Swazi
SouthAfrica Venda
SouthAfrica Tsonga
Zambia Chewa
SouthAfrica Sotho
Zambia Nkoya
Zambia Kwangwa
Zambia Mbunda
DRC Pende
Zambia Nyengo
DRC Lwer
Zambia Kwamashi
−0.0122
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−0.0116
f3
B
Sakalava Tsimihety
40°N
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20°W
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20°E
40°E
f3
0.004
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−0.004
−0.008
−0.012
Linguistic group
Bantu
Non−Bantu


A
Temoro
Mozambique Ronga
Mozambique Tswa
Mozambique Chopi
Swaziland Swazi
SouthAfrica Venda
SouthAfrica Tsonga
Zambia Chewa
Zambia Kwangwa
SouthAfrica Sotho
Zambia Nyengo
Zambia Mbunda
DRC Pende
SouthAfrica Pedi
Zambia TongaZam
Zambia Nkoya
−0.0120
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−0.0116
−0.0114
f3
B
Temoro
40°N
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40°E
f3
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0.000
−0.004
−0.008
−0.012
Linguistic group
Bantu
Non−Bantu




```
bet_all_afr <- res_afr[[1]]
```

## Add Bantu language group

```
matching <- match(
  mad_bor_g1000_sum_ld_no_na$population,
  linguistic_groups$`Population label`
)
mad_bor_g1000_sum_ld_no_na <- mad_bor_g1000_sum_ld_no_na %>%
  mutate(bantu_group = linguistic_groups$`Bantu Group`[matching])

mad_bor_g1000_sum_ld_no_na$bantu_group <-
  ifelse(mad_bor_g1000_sum_ld_no_na$population == "Zambia_Tumbuka",
    "E-BSP",
    mad_bor_g1000_sum_ld_no_na$bantu_group
  )
mad_bor_g1000_sum_ld_no_na$bantu_group <-
  ifelse(mad_bor_g1000_sum_ld_no_na$population == "LWK",
    "E-BSP",
    mad_bor_g1000_sum_ld_no_na$bantu_group
  )

# Bantu groups:
```

```
# E-BSP: Eastern Bantu-speaking populations
# NW-BSP 2: North-Western Bantu-speaking populations
# SW-BSP: South-Western Bantu-speaking populations
# W-BSP: West-Western Bantu-speaking populations

# replace bantu_group acronym with full name
mad_bor_g1000_sum_ld_no_na$bantu_group <-
  ifelse(mad_bor_g1000_sum_ld_no_na$bantu_group == "E-BSP",
    "Eastern Bantu",
    mad_bor_g1000_sum_ld_no_na$bantu_group
  )
mad_bor_g1000_sum_ld_no_na$bantu_group <-
  ifelse(mad_bor_g1000_sum_ld_no_na$bantu_group == "NW-BSP 2",
    "North-Western Bantu",
    mad_bor_g1000_sum_ld_no_na$bantu_group
  )
mad_bor_g1000_sum_ld_no_na$bantu_group <-
  ifelse(mad_bor_g1000_sum_ld_no_na$bantu_group == "SW-BSP",
    "South-Western Bantu",
    mad_bor_g1000_sum_ld_no_na$bantu_group
  )
mad_bor_g1000_sum_ld_no_na$bantu_group <-
  ifelse(mad_bor_g1000_sum_ld_no_na$bantu_group == "W-BSP",
    "Western Bantu",
    mad_bor_g1000_sum_ld_no_na$bantu_group
  )
```

## Plot f3's on the map: Bantu groups

```
plot_bantu_maps <- function(subset_list, mad_bor_g1000_sum_ld_no_na, africa) {
  counter <- 1
  plot_list <- list()

  for (i in subset_list) {
    malagasy_pop <- str_replace(i, "adm_f3_african_", "")
    f3 <- paste0("f3_", malagasy_pop)
    se <- paste0("se_", malagasy_pop)

    mad_bantu <- mad_bor_g1000_sum_ld_no_na %>%
      filter(!bantu_group == "-")

    # minimum f3
    minimum_f3 <- mad_bantu %>%
      slice_min(.data[[f3]])
    minval <- unique(as.numeric(minimum_f3[[f3]]))

    upper <- minval + (10 * unique(as.numeric(minimum_f3[[se]])))

    # maximum f3
    maximum_f3 <- mad_bor_g1000_sum_ld_no_na %>%
      slice_max(.data[[f3]])
```

```
    maxval <- unique(as.numeric(maximum_f3[[f3]]))

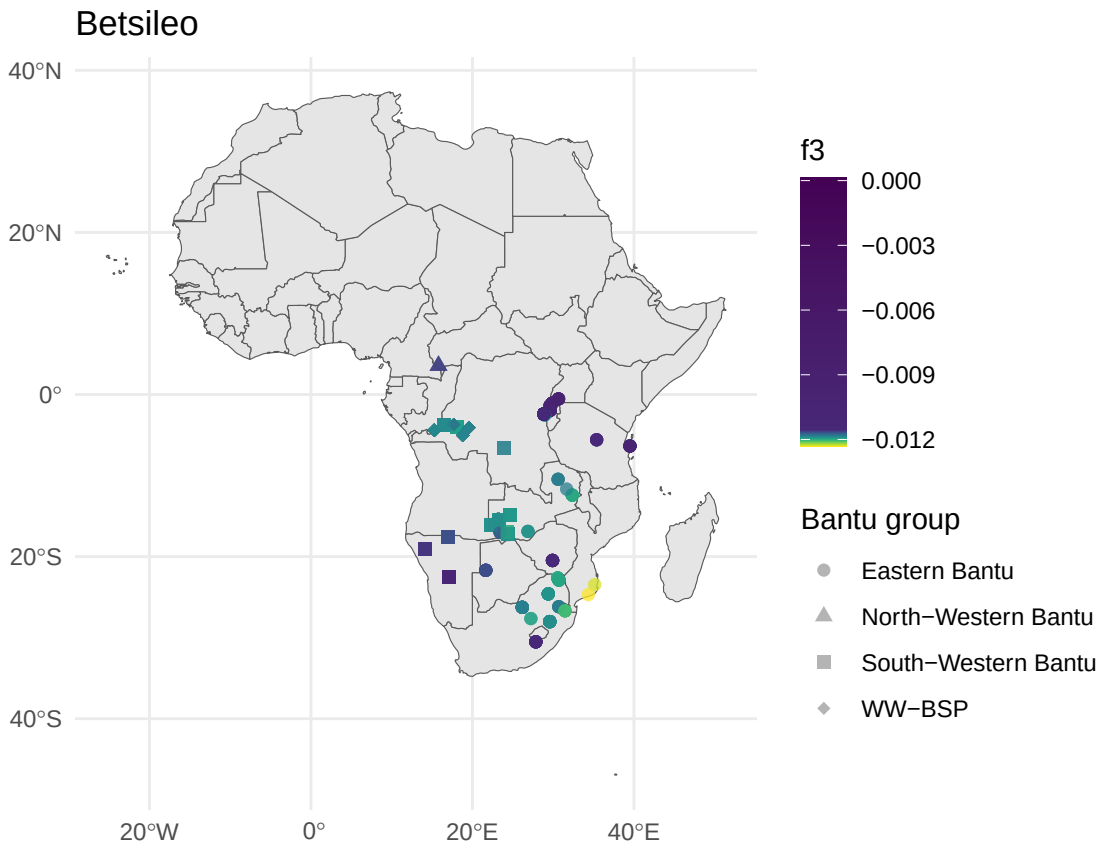
    this_plot <- ggplot() +
      geom_sf(data = africa) +
      geom_sf(
        data = mad_bantu,
        aes(colour = .data[[f3]], shape = bantu_group),
        alpha = 0.3, size = 2
      ) +
      scale_color_gradientn(
        colors = rev(viridis::viridis(10)),
        na.value = "grey80",
        limits = c(minval, maxval), # Using the calculated minval
        # Mapping: Purple at min, Yellow at -0.009, End of scale at 0
        values = scales::rescale(c(seq(minval, upper, length.out = 9), maxval)),
        guide = guide_colorbar(barheight = 7),
        oob = scales::squish
      ) +
      # make shapes all filled and not hollow
      scale_shape_manual(values = c(16, 17, 15, 18, 19)) +
      coord_sf(crs = st_crs(4326)) +
      theme_minimal() +
      labs(colour = "f3", shape = "Bantu group") +
      guides(alpha = "none", size = "none") +
      ggtitle(malagasy_pop)

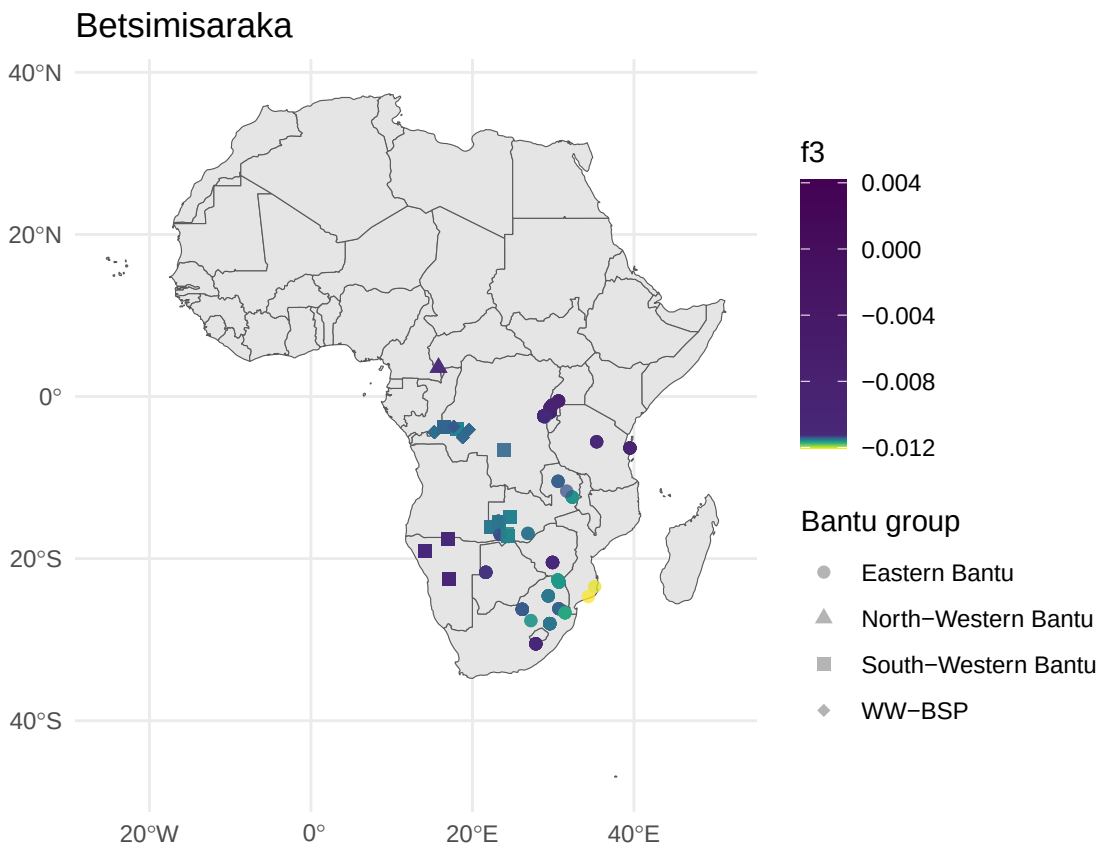
    plot_list[[counter]] <- this_plot
    counter <- counter + 1
  }

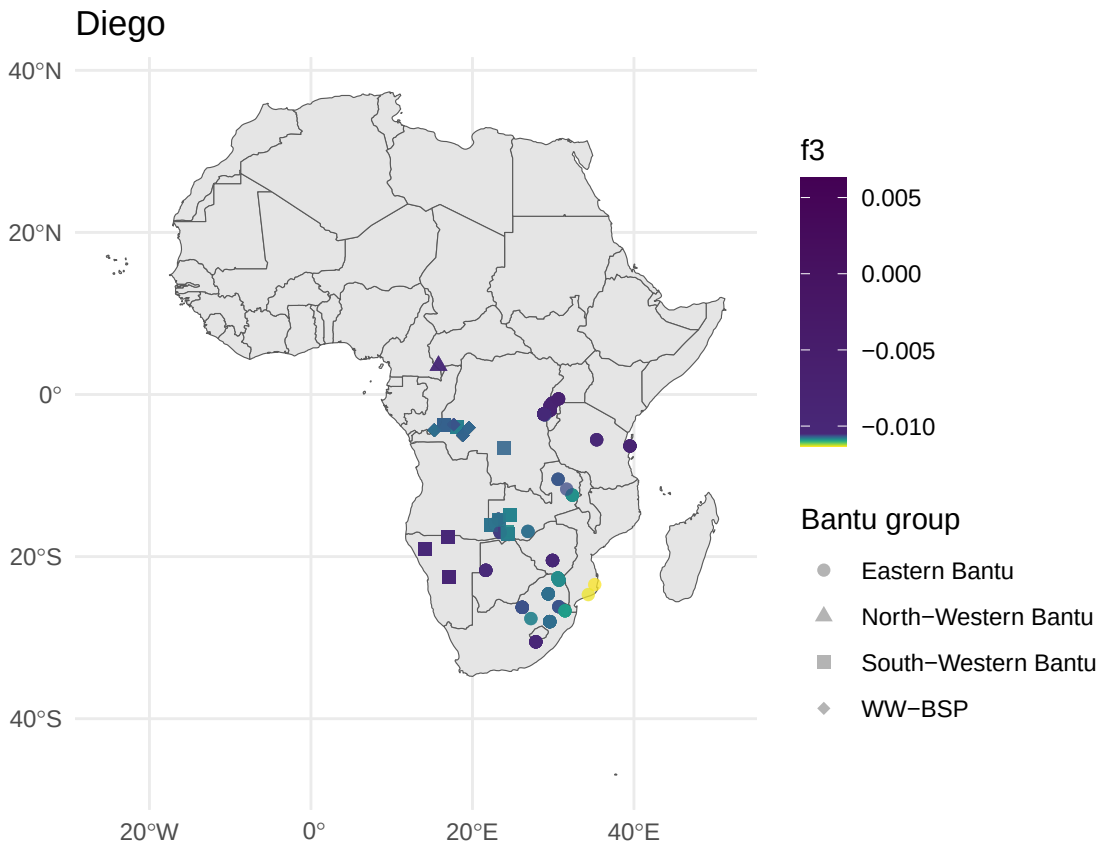
  return(plot_list)
}

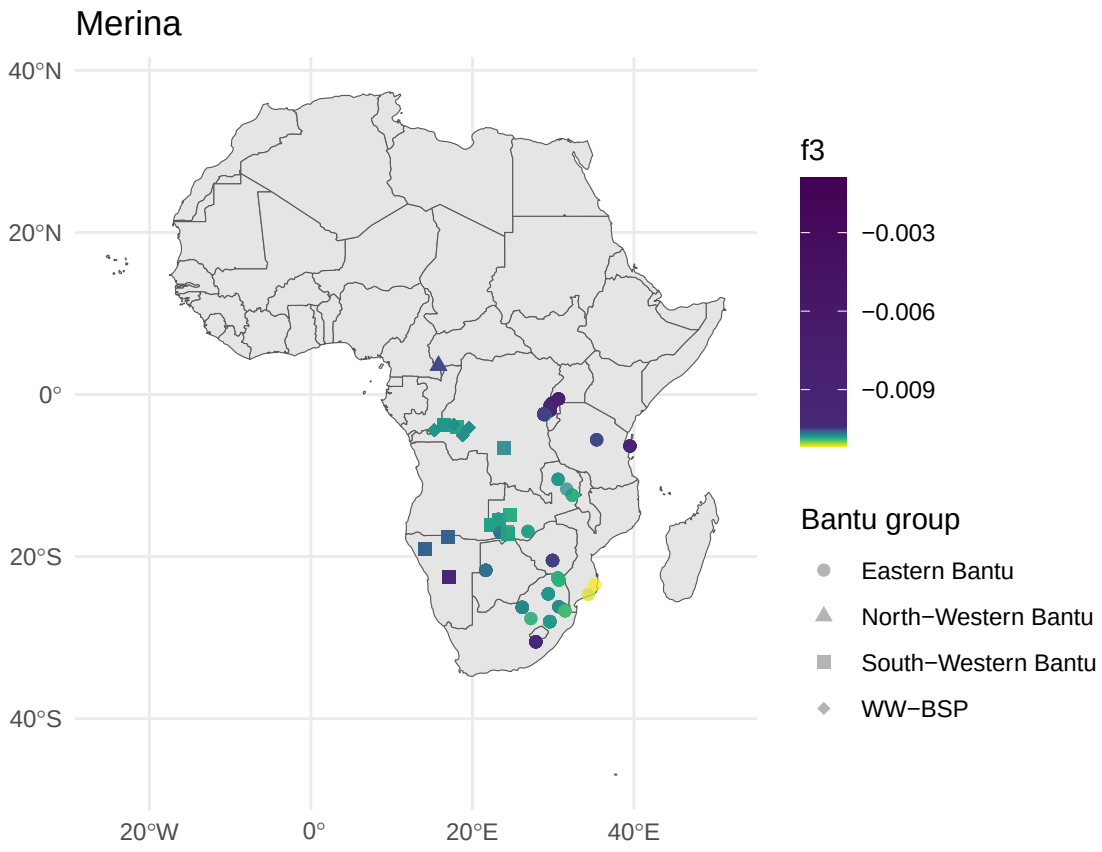
res_bantu <- plot_bantu_maps(subset_list, mad_bor_g1000_sum_ld_no_na, africa)

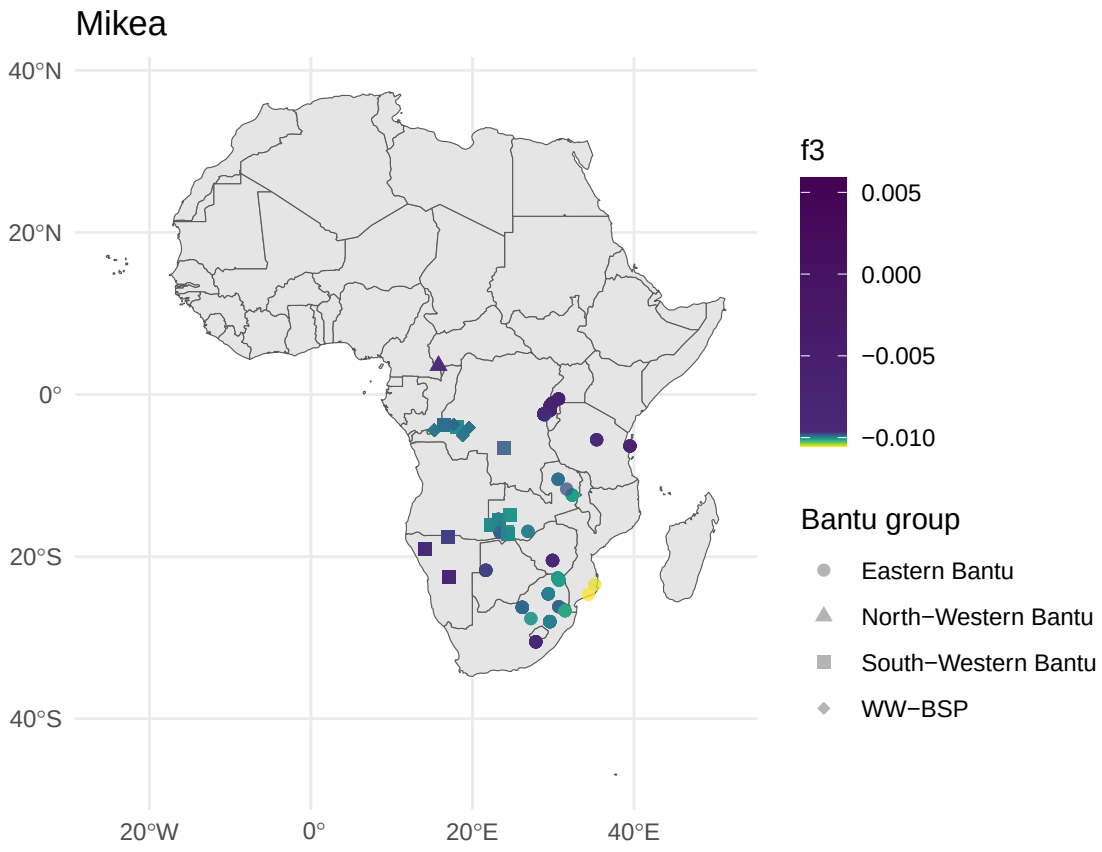
for (p in res_bantu) {
  print(p)
  ggsave(paste0(res_dir, "f3_bantu_pops_", p$labels$title, ".jpg"),
    width = 6, height = 4, dpi = 300
  )
}
```

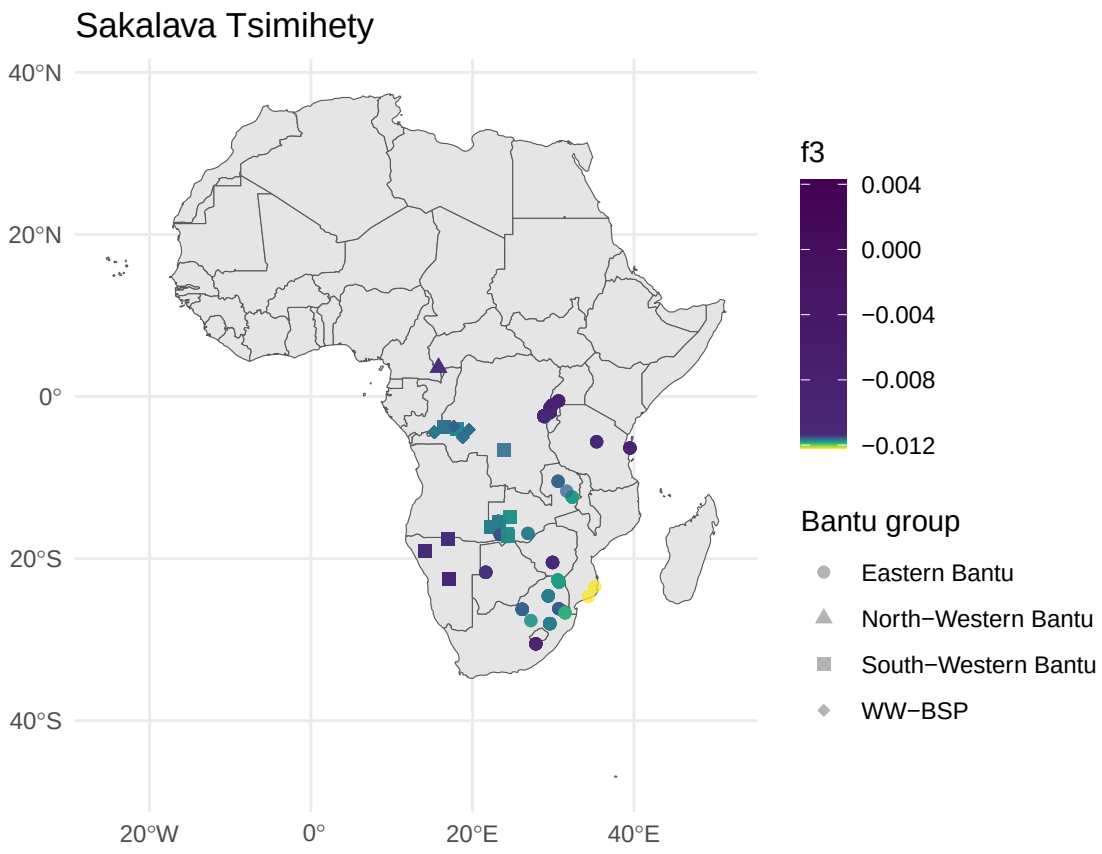

Betsileo
40°N
20°N
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20°S
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0°
20°E
40°E
f3
0.000
−0.003
−0.006
−0.009
−0.012
Bantu group
Eastern Bantu
North−Western Bantu
South−Western Bantu
WW−BSP

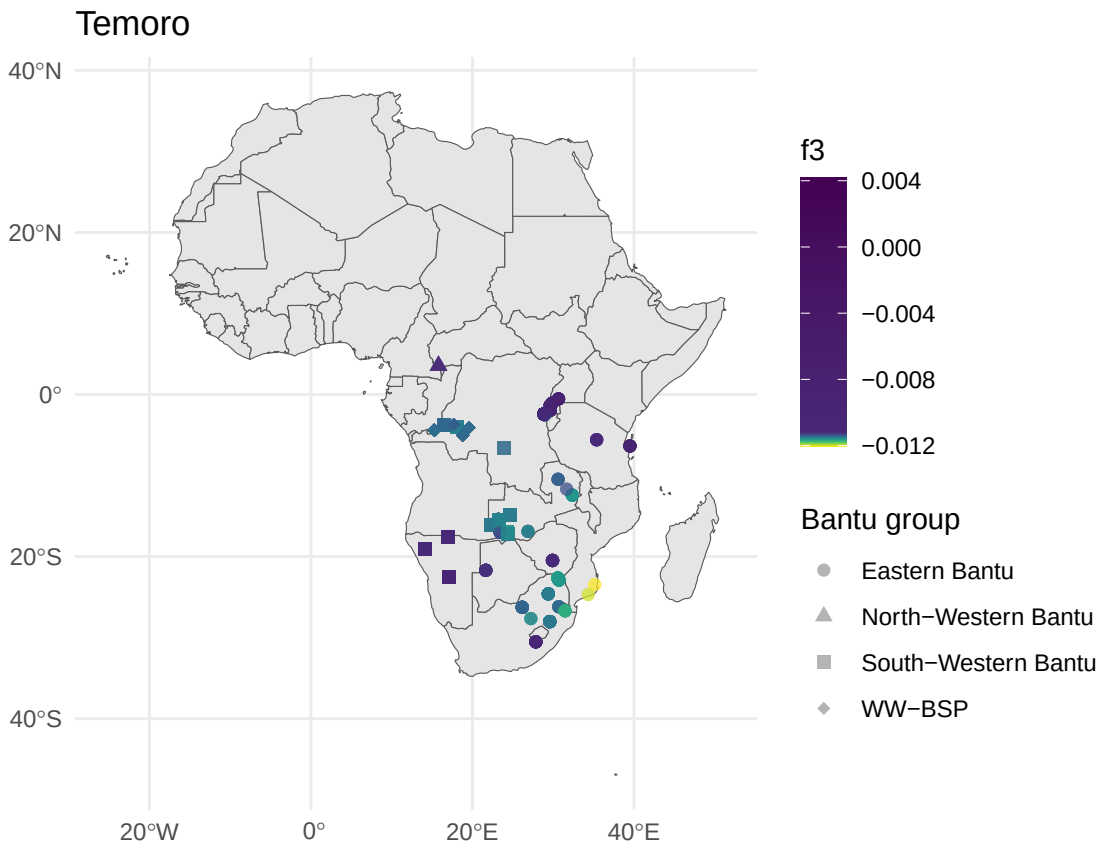

Betsimisaraka
40°N
20°N
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40°S
20°W
0°
20°E
40°E
f3
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−0.012
Bantu group
Eastern Bantu
North−Western Bantu
South−Western Bantu
WW−BSP

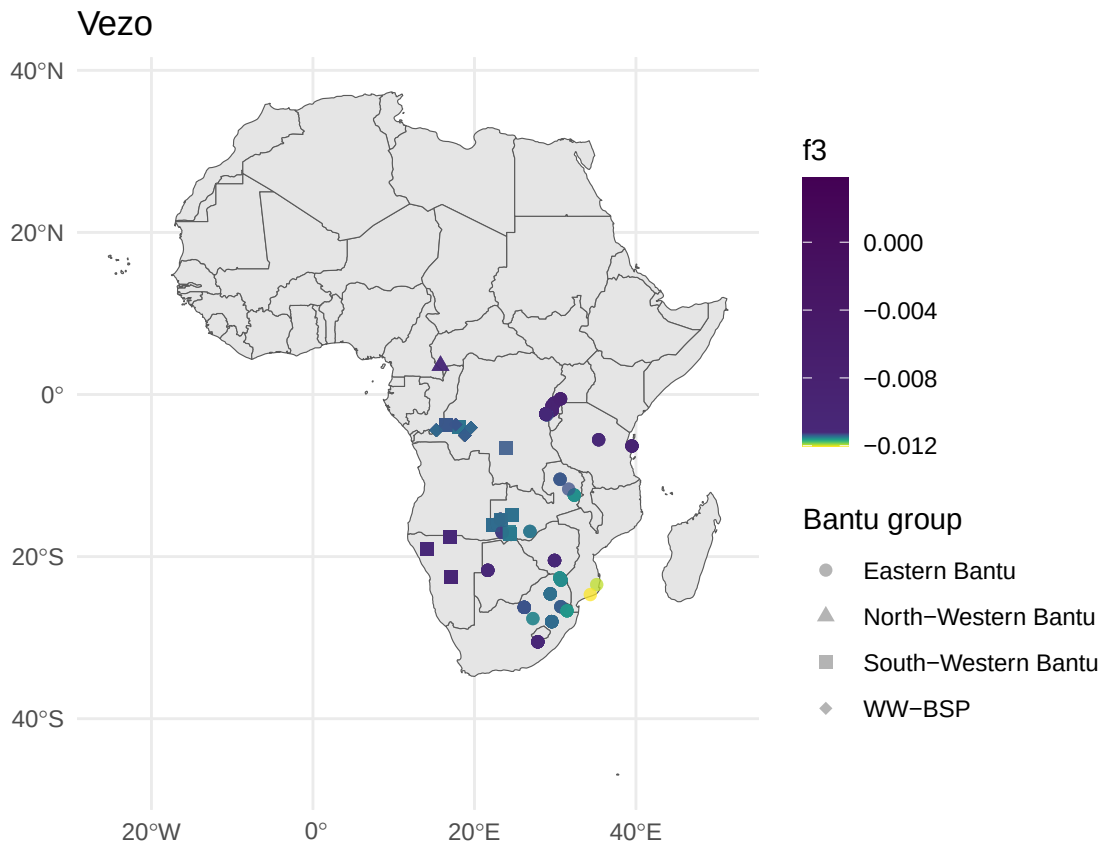

Diego
40°N
20°N
0°
20°S
40°S
20°W
0°
20°E
40°E
f3
0.005
0.000
−0.005
−0.010
Bantu group
Eastern Bantu
North−Western Bantu
South−Western Bantu
WW−BSP


Merina
40°N
20°N
0°
20°S
40°S
20°W
0°
20°E
40°E
f3
−0.003
−0.006
−0.009
Bantu group
Eastern Bantu
North−Western Bantu
South−Western Bantu
WW−BSP


Mikea
40°N
20°N
0°
20°S
40°S
20°W
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40°E
f3
0.005
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−0.005
−0.010
Bantu group
Eastern Bantu
North−Western Bantu
South−Western Bantu
WW−BSP


Sakalava Tsimihety
40°N
20°N
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20°S
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20°W
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20°E
40°E
f3
0.004
0.000
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−0.008
−0.012
Bantu group
Eastern Bantu
North−Western Bantu
South−Western Bantu
WW−BSP


Temoro
40°N
20°N
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20°S
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f3
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−0.012
Bantu group
Eastern Bantu
North−Western Bantu
South−Western Bantu
WW−BSP

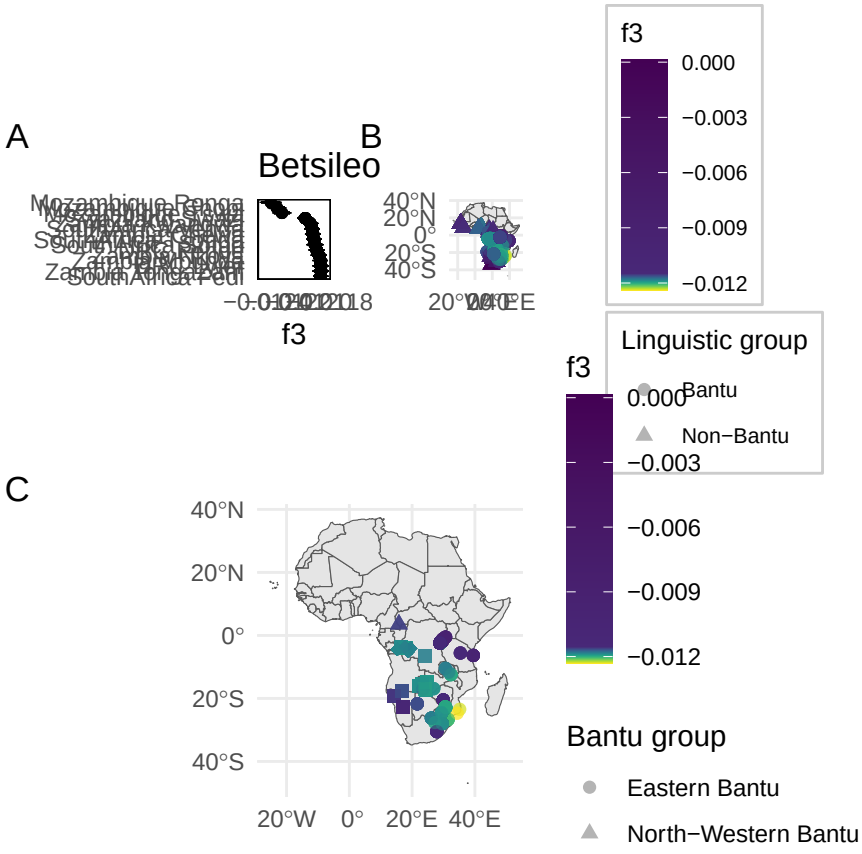



```
bet_bantu_afr <- res_bantu[[1]]
```

## Figure 2

```
res_afr[[1]]$labels$title <- NULL
res_bantu[[1]]$labels$title <- NULL

combined_plot <- wrap_plots(c(res_afr[1], res_bantu[1]), nrow = 2) +
  plot_annotation(
    tag_levels = "A" # Automatically labels them A, B, C...
  )
combined_plot
```




```
ggsave(paste0(res_dir, "f3_combined_plot.jpg"),
  width = 8, height = 10, dpi = 300
)
```

---